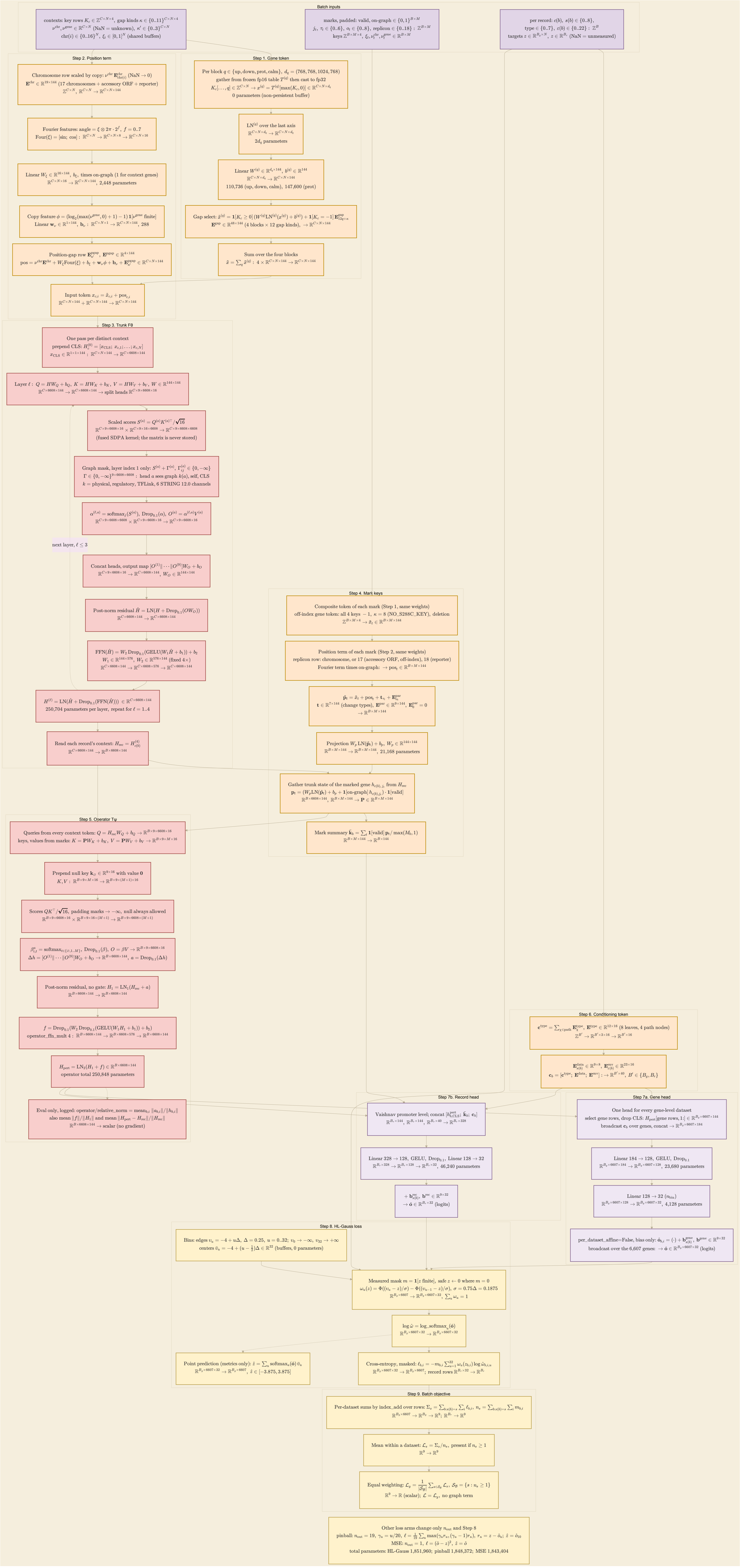

Batch inputs

contexts: key rows $K_c \in \mathbb{Z}^{C\times N\times 4}$, gap kinds $\kappa \in \{0..11\}^{C\times N\times 4}$
$\nu^{\text{chr}}, \nu^{\text{gene}} \in \mathbb{R}^{C\times N}$ (NaN = unknown), $\kappa' \in \{0..3\}^{C\times N}$
$\text{chr}(i) \in \{0..16\}^N$, $\xi_i \in [0,1]^N$ (shared buffers)

marks, padded: valid, on-graph $\in \{0,1\}^{B\times M}$
$j_t,\ \tau_t \in \{0..6\},\ o_t \in \{0..8\}$, replicon $\in \{0..18\} :\ \mathbb{Z}^{B\times M}$
keys $\mathbb{Z}^{B\times M\times 4},\ \xi_t, \nu_t^{\text{chr}}, \nu_t^{\text{gene}} \in \mathbb{R}^{B\times M}$

per record: $c(b),\ s(b) \in \{0..8\}$,
type $\in \{0..7\},\ \varepsilon(b) \in \{0..22\} :\ \mathbb{Z}^B$
targets $z \in \mathbb{R}^{B_g\times N},\ z \in \mathbb{R}^{B_r}$ (NaN = unmeasured)

Step 2. Position term

Chromosome row scaled by copy: $\nu^{\text{chr}}\,\mathbf{E}^{\text{chr}}_{\text{chr}(i)}$ (NaN $\to 0$)
$\mathbf{E}^{\text{chr}} \in \mathbb{R}^{19\times 144}$ (17 chromosomes + accessory ORF + reporter)
$\mathbb{Z}^{C\times N},\ \mathbb{R}^{C\times N} \to \mathbb{R}^{C\times N\times 144}$

Fourier features: angle $= \xi \otimes 2\pi\cdot 2^f,\ f = 0..7$
$\text{Four}(\xi) = [\sin;\ \cos] :\ \mathbb{R}^{C\times N} \to \mathbb{R}^{C\times N\times 8} \to \mathbb{R}^{C\times N\times 16}$

Linear $W_\xi \in \mathbb{R}^{16\times 144},\ b_\xi$, times on-graph (1 for context genes)
$\mathbb{R}^{C\times N\times 16} \to \mathbb{R}^{C\times N\times 144}$, 2,448 parameters

Copy feature $\phi = (\log_2(\max(\nu^{\text{gene}}, 0) + 1) - 1)\,\mathbf{1}[\nu^{\text{gene}}\ \text{finite}]$
Linear $\mathbf{w}_\nu \in \mathbb{R}^{1\times 144},\ \mathbf{b}_\nu :\ \mathbb{R}^{C\times N\times 1} \to \mathbb{R}^{C\times N\times 144}$, 288

Position-gap row $\mathbf{E}^{\text{pgap}}_{\kappa'},\ \mathbf{E}^{\text{pgap}} \in \mathbb{R}^{4\times 144}$
$\text{pos} = \nu^{\text{chr}}\mathbf{E}^{\text{chr}} + W_\xi\text{Four}(\xi) + b_\xi + \mathbf{w}_\nu\phi + \mathbf{b}_\nu + \mathbf{E}^{\text{pgap}}_{\kappa'} \in \mathbb{R}^{C\times N\times 144}$

Input token $x_{c,i} = \tilde{x}_{c,i} + \text{pos}_{c,i}$
$\mathbb{R}^{C\times N\times 144} + \mathbb{R}^{C\times N\times 144} \to \mathbb{R}^{C\times N\times 144}$

Step 1. Gene token

Per block $q \in \{\text{up}, \text{down}, \text{prot}, \text{calm}\},\ d_q = (768, 768, 1024, 768)$
gather from frozen fp16 table $T^{(q)}$ then cast to fp32
$K_c[\ldots, q] \in \mathbb{Z}^{C\times N} \to x^{(q)} = T^{(q)}[\max(K_c, 0)] \in \mathbb{R}^{C\times N\times d_q}$
0 parameters (non-persistent buffer)

$\text{LN}^{(q)}$ over the last axis
$\mathbb{R}^{C\times N\times d_q} \to \mathbb{R}^{C\times N\times d_q}$
$2d_q$ parameters

Linear $W^{(q)} \in \mathbb{R}^{d_q\times 144},\ b^{(q)} \in \mathbb{R}^{144}$
$\mathbb{R}^{C\times N\times d_q} \to \mathbb{R}^{C\times N\times 144}$
110,736 (up, down, calm), 147,600 (prot)

Gap select: $\tilde{x}^{(q)} = \mathbf{1}[K_c \geq 0]\,(W^{(q)}\text{LN}^{(q)}(x^{(q)}) + b^{(q)}) + \mathbf{1}[K_c = -1]\,\mathbf{E}^{\text{gap}}_{12q+\kappa}$
$\mathbf{E}^{\text{gap}} \in \mathbb{R}^{48\times 144}$ (4 blocks × 12 gap kinds), $\to \mathbb{R}^{C\times N\times 144}$

Sum over the four blocks
$\tilde{x} = \sum_q \tilde{x}^{(q)} :\ 4 \times \mathbb{R}^{C\times N\times 144} \to \mathbb{R}^{C\times N\times 144}$

Step 3. Trunk Fθ

One pass per distinct context
prepend CLS: $H_c^{(0)} = [x_{\text{CLS}};\ x_{c,1}; \ldots; x_{c,N}]$
$x_{\text{CLS}} \in \mathbb{R}^{1\times 1\times 144} :\ \mathbb{R}^{C\times N\times 144} \to \mathbb{R}^{C\times 6608\times 144}$

Layer $\ell :\ Q = HW_Q + b_Q,\ K = HW_K + b_K,\ V = HW_V + b_V,\ W \in \mathbb{R}^{144\times 144}$
$\mathbb{R}^{C\times 6608\times 144} \to \mathbb{R}^{C\times 6608\times 144} \to$ split heads $\mathbb{R}^{C\times 9\times 6608\times 16}$

Scaled scores $S^{(a)} = Q^{(a)}K^{(a)\top}/\sqrt{16}$
$\mathbb{R}^{C\times 9\times 6608\times 16} \times \mathbb{R}^{C\times 9\times 16\times 6608} \to \mathbb{R}^{C\times 9\times 6608\times 6608}$
(fused SDPA kernel; the matrix is never stored)

Graph mask, layer index 1 only: $S^{(a)} + \Gamma^{(a)},\ \Gamma^{(a)}_{ij} \in \{0, -\infty\}$
$\Gamma \in \{0, -\infty\}^{9\times 6608\times 6608}$: head a sees graph $k(a)$, self, CLS
k = physical, regulatory, TFLink, 6 STRING 12.0 channels

$\alpha^{(\ell,a)} = \text{softmax}_j(S^{(a)}),\ \text{Drop}_{0.1}(\alpha),\ O^{(a)} = \alpha^{(\ell,a)}V^{(a)}$
$\mathbb{R}^{C\times 9\times 6608\times 6608} \times \mathbb{R}^{C\times 9\times 6608\times 16} \to \mathbb{R}^{C\times 9\times 6608\times 16}$

next layer, $\ell \leq 3$

Concat heads, output map $[O^{(1)}\|\cdots\|O^{(9)}]W_O + b_O$
$\mathbb{R}^{C\times 9\times 6608\times 16} \to \mathbb{R}^{C\times 6608\times 144},\ W_O \in \mathbb{R}^{144\times 144}$

Post-norm residual $\hat{H} = \text{LN}(H + \text{Drop}_{0.1}(OW_O))$
$\mathbb{R}^{C\times 6608\times 144} \to \mathbb{R}^{C\times 6608\times 144}$

$\text{FFN}(\hat{H}) = W_2\,\text{Drop}_{0.1}(\text{GELU}(W_1\hat{H} + b_1)) + b_2$
$W_1 \in \mathbb{R}^{144\times 576},\ W_2 \in \mathbb{R}^{576\times 144}$ (fixed 4×)
$\mathbb{R}^{C\times 6608\times 144} \to \mathbb{R}^{C\times 6608\times 576} \to \mathbb{R}^{C\times 6608\times 144}$

$H^{(\ell)} = \text{LN}(\hat{H} + \text{Drop}_{0.1}(\text{FFN}(\hat{H}))) \in \mathbb{R}^{C\times 6608\times 144}$
250,704 parameters per layer, repeat for $\ell = 1..4$

Read each record's context: $H_{\text{rec}} = H^{(4)}_{c(b)}$
$\mathbb{R}^{C\times 6608\times 144} \to \mathbb{R}^{B\times 6608\times 144}$

Step 4. Mark keys

Composite token of each mark (Step 1, same weights)
off-index gene token: all 4 keys $-1,\ \kappa = 8$ (NO_S288C_KEY), deletion
$\mathbb{Z}^{B\times M\times 4} \to \tilde{x}_t \in \mathbb{R}^{B\times M\times 144}$

Position term of each mark (Step 2, same weights)
replicon row: chromosome, or 17 (accessory ORF, off-index), 18 (reporter)
Fourier term times on-graph: $\to \text{pos}_t \in \mathbb{R}^{B\times M\times 144}$

$\tilde{\mathbf{p}}_t = \tilde{x}_t + \text{pos}_t + \mathbf{t}_{\tau_t} + \mathbf{E}^{\text{par}}_{o_t}$
$\mathbf{t} \in \mathbb{R}^{7\times 144}$ (change types), $\mathbf{E}^{\text{par}} \in \mathbb{R}^{9\times 144},\ \mathbf{E}^{\text{par}}_0 = 0$
$\to \mathbb{R}^{B\times M\times 144}$

Projection $W_p\,\text{LN}(\tilde{\mathbf{p}}_t) + b_p,\ W_p \in \mathbb{R}^{144\times 144}$
$\mathbb{R}^{B\times M\times 144} \to \mathbb{R}^{B\times M\times 144}$, 21,168 parameters

Gather trunk state of the marked gene $h_{c(b),j_t}$ from H_{rec}
$\mathbf{p}_t = (W_p\text{LN}(\tilde{\mathbf{p}}_t) + b_p + \mathbf{1}[\text{on-graph}]\,h_{c(b),j_t}) \cdot \mathbf{1}[\text{valid}]$
$\mathbb{R}^{B\times 6608\times 144},\ \mathbb{R}^{B\times M\times 144} \to \mathbf{P} \in \mathbb{R}^{B\times M\times 144}$

Mark summary $\bar{\mathbf{k}}_b = \sum_t \mathbf{1}[\text{valid}]\,\mathbf{p}_t/\max(M_b, 1)$
$\mathbb{R}^{B\times M\times 144} \to \mathbb{R}^{B\times 144}$

Step 5. Operator Tψ

Queries from every context token: $Q = H_{\text{rec}}W_Q + b_Q \to \mathbb{R}^{B\times 9\times 6608\times 16}$
keys, values from marks: $K = \mathbf{P}W_K + b_K,\ V = \mathbf{P}W_V + b_V \to \mathbb{R}^{B\times 9\times M\times 16}$

Prepend null key $\mathbf{k}_\varnothing \in \mathbb{R}^{9\times 16}$ with value $\mathbf{0}$
$K, V :\ \mathbb{R}^{B\times 9\times M\times 16} \to \mathbb{R}^{B\times 9\times (M+1)\times 16}$

Scores $QK^\top/\sqrt{16}$, padding marks $\to -\infty$, null always allowed
$\mathbb{R}^{B\times 9\times 6608\times 16} \times \mathbb{R}^{B\times 9\times 16\times (M+1)} \to \mathbb{R}^{B\times 9\times 6608\times (M+1)}$

$\beta^a_{i,t} = \text{softmax}_{t\in\{\varnothing,1..M\}},\ \text{Drop}_{0.1}(\beta),\ O = \beta V \to \mathbb{R}^{B\times 9\times 6608\times 16}$
$\Delta h = [O^{(1)}\|\cdots\|O^{(9)}]W_O + b_O \to \mathbb{R}^{B\times 6608\times 144},\ a = \text{Drop}_{0.1}(\Delta h)$

Post-norm residual, no gate: $H_1 = \text{LN}_1(H_{\text{rec}} + a)$
$\mathbb{R}^{B\times 6608\times 144} \to \mathbb{R}^{B\times 6608\times 144}$

$f = \text{Drop}_{0.1}(W_2\,\text{Drop}_{0.1}(\text{GELU}(W_1H_1 + b_1)) + b_2)$
operator_ffn_mult 4 : $\mathbb{R}^{B\times 6608\times 144} \to \mathbb{R}^{B\times 6608\times 576} \to \mathbb{R}^{B\times 6608\times 144}$

$H_{\text{pert}} = \text{LN}_2(H_1 + f) \in \mathbb{R}^{B\times 6608\times 144}$
operator total 250,848 parameters

Eval only, logged: operator/relative_norm $= \text{mean}_{b,i}\,\|a_{b,i}\|/\|h_{b,i}\|$
also mean $\|f\|/\|H_1\|$ and mean $\|H_{\text{pert}} - H_{\text{rec}}\|/\|H_{\text{rec}}\|$
$\mathbb{R}^{B\times 6608\times 144} \to$ scalar (no gradient)

Step 6. Conditioning token

$\mathbf{c}^{\text{type}} = \sum_{\chi\in\text{path}} \mathbf{E}^{\text{type}}_\chi,\ \mathbf{E}^{\text{type}} \in \mathbb{R}^{12\times 16}$ (8 leaves, 4 path nodes)
$\mathbb{Z}^{B'} \to \mathbb{R}^{B'\times 3\times 16} \to \mathbb{R}^{B'\times 16}$

$\mathbf{E}^{\text{data}}_{s(b)} \in \mathbb{R}^{9\times 8},\ \mathbf{E}^{\text{env}}_{\varepsilon(b)} \in \mathbb{R}^{23\times 16}$
$\mathbf{c}_b = [\mathbf{c}^{\text{type}};\ \mathbf{E}^{\text{data}};\ \mathbf{E}^{\text{env}}] : \to \mathbb{R}^{B'\times 40},\ B' \in \{B_g, B_r\}$

Step 7b. Record head

Vaishnav promoter level; concat $[h^{\text{pert}}_{b,\text{CLS}};\ \bar{\mathbf{k}}_b;\ \mathbf{c}_b]$
$\mathbb{R}^{B_r\times 144},\ \mathbb{R}^{B_r\times 144},\ \mathbb{R}^{B_r\times 40} \to \mathbb{R}^{B_r\times 328}$

Linear $328 \to 128$, GELU, $\text{Drop}_{0.1}$, Linear $128 \to 32$
$\mathbb{R}^{B_r\times 328} \to \mathbb{R}^{B_r\times 128} \to \mathbb{R}^{B_r\times 32}$, 46,240 parameters

$+\,\mathbf{b}^{\text{rec}}_{s(b)},\ \mathbf{b}^{\text{rec}} \in \mathbb{R}^{9\times 32}$
$\to \hat{\mathbf{o}} \in \mathbb{R}^{B_r\times 32}$ (logits)

Step 7a. Gene head

One head for every gene-level dataset
select gene rows, drop CLS: $H_{\text{pert}}[\text{gene rows}, 1{:}] \in \mathbb{R}^{B_g\times 6607\times 144}$
broadcast $\mathbf{c}_b$ over genes, concat $\to \mathbb{R}^{B_g\times 6607\times 184}$

Linear $184 \to 128$, GELU, $\text{Drop}_{0.1}$
$\mathbb{R}^{B_g\times 6607\times 184} \to \mathbb{R}^{B_g\times 6607\times 128}$, 23,680 parameters

Linear $128 \to 32$ (n_{bin})
$\mathbb{R}^{B_g\times 6607\times 128} \to \mathbb{R}^{B_g\times 6607\times 32}$, 4,128 parameters

per_dataset_affine=False, bias only: $\hat{\mathbf{o}}_{b,i} = (\cdot) + \mathbf{b}^{\text{gene}}_{s(b)},\ \mathbf{b}^{\text{gene}} \in \mathbb{R}^{9\times 32}$
broadcast over the 6,607 genes: $\to \hat{\mathbf{o}} \in \mathbb{R}^{B_g\times 6607\times 32}$ (logits)

Step 8. HL-Gauss loss

Bins: edges $v_u = -4 + u\Delta,\ \Delta = 0.25,\ u = 0..32;\ v_0 \to -\infty,\ v_{32} \to +\infty$
centers $\bar{v}_u = -4 + (u - \frac{1}{2})\Delta \in \mathbb{R}^{32}$ (buffers, 0 parameters)

Measured mask $m = \mathbf{1}[z\ \text{finite}]$, safe $z \leftarrow 0$ where $m = 0$
$\omega_u(z) = \Phi((v_u - z)/\sigma) - \Phi((v_{u-1} - z)/\sigma),\ \sigma = 0.75\Delta = 0.1875$
$\mathbb{R}^{B_g\times 6607} \to \mathbb{R}^{B_g\times 6607\times 32},\ \sum_u \omega_u = 1$

$\log\hat{\omega} = \text{log\_softmax}_u(\hat{\mathbf{o}})$
$\mathbb{R}^{B_g\times 6607\times 32} \to \mathbb{R}^{B_g\times 6607\times 32}$

Point prediction (metrics only): $\hat{z} = \sum_u \text{softmax}_u(\hat{\mathbf{o}})\,\bar{v}_u$
$\mathbb{R}^{B_g\times 6607\times 32} \to \mathbb{R}^{B_g\times 6607},\ \hat{z} \in [-3.875, 3.875]$

Cross-entropy, masked: $\ell_{b,i} = -m_{b,i}\sum_{u=1}^{32} \omega_u(z_{b,i})\log\hat{\omega}_{b,i,u}$
$\mathbb{R}^{B_g\times 6607\times 32} \to \mathbb{R}^{B_g\times 6607}$; record rows $\mathbb{R}^{B_r\times 32} \to \mathbb{R}^{B_r}$

Step 9. Batch objective

Per-dataset sums by index_add over rows: $\Sigma_s = \sum_{b:s(b)=s}\sum_i \ell_{b,i},\ n_s = \sum_{b:s(b)=s}\sum_i m_{b,i}$
$\mathbb{R}^{B_g\times 6607} \to \mathbb{R}^{B_g} \to \mathbb{R}^9;\ \mathbb{R}^{B_r} \to \mathbb{R}^9$

Mean within a dataset: $\mathcal{L}_s = \Sigma_s/n_s$, present if $n_s \geq 1$
$\mathbb{R}^9 \to \mathbb{R}^9$

Equal weighting: $\mathcal{L}_y = \dfrac{1}{|\mathcal{S}_\mathcal{B}|}\sum_{s\in\mathcal{S}_\mathcal{B}} \mathcal{L}_s,\ \mathcal{S}_\mathcal{B} = \{s : n_s \geq 1\}$
$\mathbb{R}^9 \to \mathbb{R}$ (scalar); $\mathcal{L} = \mathcal{L}_y$, no graph term

Other loss arms change only n_{out} and Step 8
pinball: $n_{\text{out}} = 19,\ \gamma_u = u/20,\ \ell = \frac{1}{19}\sum_u \max(\gamma_u r_u, (\gamma_u - 1)r_u),\ r_u = z - \hat{o}_u;\ \hat{z} = \hat{o}_{10}$
MSE: $n_{\text{out}} = 1,\ \ell = (\hat{o} - z)^2,\ \hat{z} = \hat{o}$
total parameters: HL-Gauss 1,851,960; pinball 1,848,372; MSE 1,843,404